

Project ID:

R26-SE-037

1. Topic (12 words max)

SinhalaJournal-LLM: A Style-Controlled Large Language Model for Diverse Sri Lankan Newspaper Writing

2. Research group the project belongs to

CoEAI - Centre of Excellence for AI

3. Specialization of the project belongs to

Software Engineering (SE)

4. If a continuation of a previous project:

Project ID	R26-SE-037
Year	2026

5. Brief description of the research problem including references (300 words max) – references not included in word count.

Sinhala journalism in Sri Lanka faces persistent challenges due to the lack of advanced Natural Language Processing (NLP) tools specifically designed for Sinhala newspaper writing. Newsrooms, digital media platforms, and journalism students continue to rely heavily on manual processes for headline creation, grammar correction, style adaptation, and summarization. These tasks demand experienced editors and significant time, leading to slow editorial workflows and inconsistent writing quality across articles. The primary groups affected by this problem include Sinhala journalists, editors, media institutions, journalism students, and online news publishers who require fast and high-quality content production.

From a technical perspective, Sinhala is considered a low-resource language, with limited large-scale, high-quality datasets and few language models capable of handling its grammar, morphology, and stylistic variations. Existing multilingual large language models perform inadequately when applied to Sinhala news writing, particularly in tasks that require style control and journalistic accuracy. Socially, inconsistent grammar, unclear summaries, and poorly written headlines reduce readability and reader trust in news content. Economically, the heavy dependence on manual editing increases operational costs and reduces the competitiveness of Sinhala media outlets in the rapidly evolving digital news environment. Ethically and legally, inaccurate or poorly structured news content may contribute to misinformation or reputational harm.

This research is positioned within the Artificial Intelligence and Data Science research cluster, focusing on the Natural Language Processing and Media Technology industry verticals. It aims to address the identified gaps by developing SinhalaJournal-LLM, a style-controlled language model tailored for Sri Lankan newspaper writing.

Furthermore, the project aligns with several United Nations Sustainable Development Goals (SDGs), including SDG 4 (Quality Education) by supporting Sinhala language learning, SDG 9 (Industry, Innovation and Infrastructure) by advancing AI-based media technologies, and SDG 16 (Peace, Justice and Strong Institutions) by promoting clear, accurate, and responsible news communication.

- Bird, S., Klein, E., & Loper, E. (2009). *Natural Language Processing with Python*. O'Reilly Media.
- Joshi, P., Santy, S., Budhiraja, A., Bali, K., & Choudhury, M. (2020). The State and Fate of Linguistic Diversity and Inclusion in the NLP World. *Proceedings of ACL*.
- Jeston, J., & Nelis, J. (2018). *Business Process Management: Practical Guidelines to Successful Implementations*. Routledge.
- UNESCO. (2021). *Journalism, 'Fake News' and Disinformation*. UNESCO Publishing.
- United Nations. (2015). *Transforming our world: the 2030 Agenda for Sustainable Development*.

6. Brief description of Existing Research and Systems (300 words max)

Summary of Existing Systems : At present, the scope of technology related to the Sri Lankan Sinhala language is restricted to general purpose apps. Google Translate, Helakuru (keyboard application), and the like support basic features like translation, typing, and so on, but are not contextually appropriate for write-ups. Some latest academic models include SinLlama (a Llama-3-based Sinhala fine-tuned model), SinMorphy (Sinhala Morphological Analyzer), but are in the prototypes phase. As far as the news sector is concerned, apps like the Automatic Sinhala News Summarizer, from the Moratuwa University, are present, which are restricted to extractive algorithms to summarize the text.

Summary of Existing Research : Research in Sinhala language processing has largely focused on basic tasks like identifying grammatical structures and specific names within text. For creative writing tasks, studies have utilized general multilingual models to generate headlines and analyze sentiment. The release of new Sinhala news datasets has enabled further research into categorizing news stories. However, most existing research on summarization still employs older, formula-based methods rather than modern, advanced AI models that can understand context and generate human-like text.

Gaps and Limitations Identified : The crucial gap is a lack of control of style. Current systems cannot detect the formal writing style necessary in editorials versus the condensed, pressing writing styles of breaking news. On another note, the "low resource" characteristic of the Sinhala language means that current multilingual systems tend to experience hallucinations in addition to an unnatural writing style. Currently, there is a gap between the world of academic prototypes versus a practical application in everyday life, as there is a lack of a system that can naturally integrate with a newsroom setting regarding grammar, summarization, and stylistic changes.

7. Brief description of the solution's nature, including a conceptual diagram (maximum 500 words).

The proposed solution, SinhalaJournal-LLM, is an AI-powered language assistance system designed to support Sinhala newspaper writing by addressing the lack of advanced, domain-specific Natural Language Processing (NLP) tools for Sinhala journalism. The system focuses on four core functionalities: headline generation, grammar correction, style rewriting, and news summarization, all tailored specifically to Sinhala news content.

Addressing Identified Gaps

As identified in Section 6, Sinhala journalism lacks high-quality datasets, grammar correction tools, and style-aware generative systems. Existing multilingual models do not adequately handle Sinhala grammar, journalistic tone, or newspaper-specific writing styles. SinhalaJournal-LLM directly addresses these gaps by (1) building a Sinhala Unicode news corpus, (2) training models specifically on Sri Lankan newspaper content, and (3) providing tools that align with real newsroom workflows. The system functions as an assistive editorial tool rather than an autonomous publisher, ensuring responsible use of AI in media.

System Perspective: Input – Process – Output

Input

- Sinhala news articles or text selected by the user
- Raw Sinhala text pasted through the web interface or Chrome plugin
- User-selected options such as target writing style or desired output type

Process

1. **Preprocessing Layer**

- Unicode normalization
- Text cleaning and segmentation

2. **Core NLP Engine (SinhalaJournal-LLM)**

- **Headline Generator:** Generates relevant Sinhala headlines from article content
- **Grammar Checker:** Detects and corrects Sinhala grammar and spelling errors
- **Style Rewriter:** Rewrites text into styles such as sports, editorial, youth, or formal while preserving meaning
- **News Summarizer:** Produces concise summaries of long Sinhala news articles

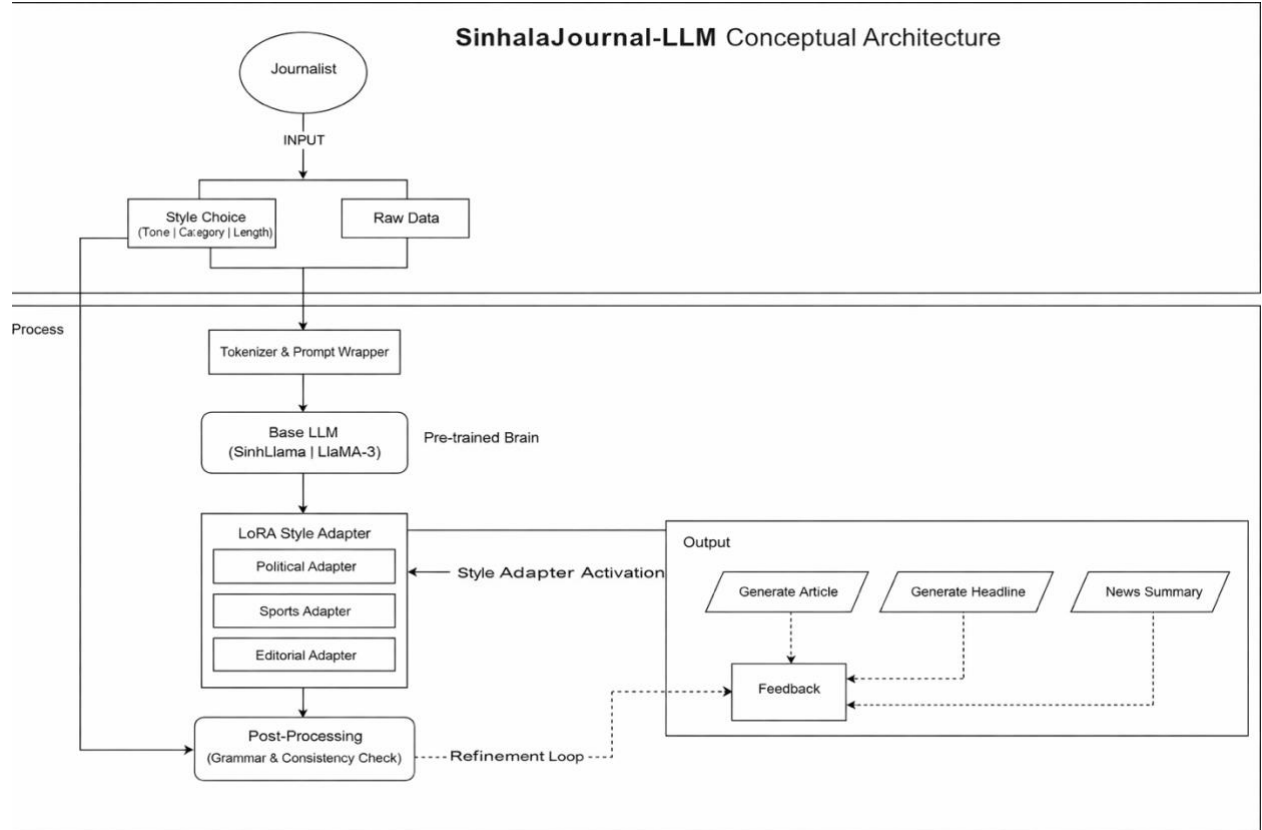
3. **Post-processing Layer**

- Output formatting
- Quality checks and readability refinement

Output

- Style-appropriate Sinhala headlines
- Grammatically corrected Sinhala text
- Rewritten content in selected newspaper styles
- Concise and informative Sinhala news summaries

These outputs help editors, journalists, students, and content creators improve efficiency, consistency, and writing quality.



8. Brief overview of the availability of necessary specialized domain expertise, knowledge, and data. (500 words max)

The project requires foundational expertise in Natural Language Processing (NLP), Large Language Models (LLMs), and computational linguistics, with specific attention to low-resource languages such as Sinhala. Core concepts related to transformer-based architectures (e.g., LLaMA) are well established and widely documented in recent academic research and open-source implementations.

In addition to technical expertise, team members can gain knowledge of journalistic writing practices, including news reporting, editorial writing, and feature article composition, through publicly documented journalism guidelines, newsroom style manuals, and analysis of existing Sinhala news publications. This combination of linguistic, technical, and journalistic knowledge enables the system to accurately model stylistic differences required in real-world newsroom environments.

The technical skills required—including Python programming, deep learning concepts, and the use of modern machine learning frameworks—align with the competencies expected from final-year undergraduate students in Information Technology. Core theoretical knowledge related to NLP, transformer-based language models, and large language model (LLM) fine-tuning is extensively documented. Concepts such as attention mechanisms, tokenization, embedding representations, and sequence-to-sequence learning are commonly covered in university curricula and research publications.

Practical knowledge related to parameter-efficient fine-tuning techniques—such as Low-Rank Adaptation (LoRA)—and retrieval-augmented generation (RAG) is readily accessible through research papers, technical blogs, and open-source implementations. These methods provide clear guidance on how large pretrained models can be adapted for domain-specific and low-resource language tasks without extensive computational requirements. Knowledge of Sinhala language processing is also expanding, with studies on morphology, syntax, and pretrained models such as SinLlama providing essential linguistic understanding. Furthermore, members can learn journalistic writing structures such as headline formulation, article composition, and stylistic conventions through direct analysis of Sinhala news content.

Despite Sinhala being considered a low-resource language, the data required for this research is sufficiently available. Large volumes of Sinhala textual data can be obtained from publicly accessible online news platforms, including Lankadeepa, Hiru News, Ada, and Divaina. These sources provide structured news articles containing headlines, full article bodies, publication dates, and category information, which are highly suitable for training and evaluating newsroom-oriented language models.

This data naturally supports multiple learning objectives, including style-aware text generation, summarization, and content validation. Because the data follows consistent journalistic formats, it enables effective supervised and semi-supervised learning for tasks such as headline generation, article continuation, and writing style adaptation.

All data sources are publicly accessible news platforms, and content usage will follow fair-use research practices. No personal or sensitive data is used, minimizing ethical concerns. Any scraping or processing of data will comply with platform terms of service and ethical research standards.

9. Objectives and Novelty

Main Objective

The main objective of this research project is to **design, develop, and evaluate a style-controlled Large Language Model (LLM) tailored for Sinhala newspaper writing**, with the aim of supporting diverse writing styles used in Sri Lankan journalism.

The project seeks to address the lack of advanced AI-based language tools for Sinhala by building a system that can understand and generate Sinhala news content in multiple newspaper styles, such as sports reporting, editorial writing, youth-oriented content, and formal news reporting. By incorporating style control mechanisms into a Large Language Model, the system enables users to generate headlines, rewrite news content in different styles, correct grammatical errors, and produce concise summaries while preserving the original meaning of the text.

A key objective of SinhalaJournal-LLM is to function as an **assistive editorial tool** for journalists, editors, and journalism students, helping them improve writing quality, maintain stylistic consistency, and reduce the time spent on repetitive editorial tasks. The system is designed to keep human users in control by allowing explicit style selection and user feedback, ensuring responsible and ethical use of AI in news writing.

Member Name with Registration No	Sub Objective	Tasks	Novelty
IT22207272	Sinhala Grammar Checker	• Study common Sinhala grammar and spelling errors in news writing • Build grammar correction rules and datasets • Train a model to detect and correct grammar errors	• One of the first Grammarly-like grammar checking systems for Sinhala • Combines linguistic rules with AI-based correction for journalistic Sinhala

		• Provide user-friendly correction suggestions	
IT22228062	Sinhala Headline Generator	• Collect Sinhala news articles with original headlines • Preprocess and normalize text into Unicode Sinhala • Train a model to generate headlines from article content • Support different headline types such as formal, sports, and breaking news	• First headline generation system focused on Sinhala newspaper writing styles • Trained on locally sourced Sri Lankan news data instead of generic multilingual datasets
IT22246332	Sinhala Style Rewriter	• Define Sinhala writing styles such as sports, editorial, youth, and formal • Train a style-controlled model to rewrite text without changing meaning • Enable users to select and apply writing styles • Evaluate rewritten text for meaning preservation	• Introduces style transfer for Sinhala news text, which is rarely addressed in existing research • Allows the same content to be rewritten into multiple newspaper styles
IT22049872	Sinhala News Summarizer	• Collect long Sinhala news articles • Design and implement a summarization approach • Train a model to generate concise summaries • Evaluate summaries for clarity and information retention	• Sinhala news summarization is under-explored • Focused specifically on news-domain summarization rather than general Sinhala text

**10. This part is to be filled by the
Supervisor and the Co-supervisor of the Project.**

- a) This research topic possesses a comprehensive scope suitable for a final-year project.

Yes	☒	No	☐
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- b) The proposed topic exhibits novelty.

Yes	☒	No	☐
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- c) The student group can successfully execute the proposed project.

Yes	☒	No	☐
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- d) The proposed sub-objectives reflect the students' areas of specialization.

Yes	☒	No	☐
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- e) Any other comments:

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	Title	First Name	Last Name	Signature
Supervisor	Prof.	Nuwan	Kodagoda	
Co-Supervisor				
External Supervisor				
Summary of external supervisor's (if any) experience and expertise				